

Conversion Insights — Feature Signals & Business Direction

Directional Analysis v1 | UNC-768 | NOT PRODUCTION-READY — Decision-Science Use Only

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1 0. Purpose and Caveats

This is a directional analysis, not a production model. N = 10 confirmed orders. We apply SMOTE to create synthetic positive examples from the real converters' nearest neighbors, then use feature importance from multiple models to identify what *consistently* separates buyers from non-buyers. Any single metric here has wide confidence intervals. The goal is hypothesis generation for marketing and product decisions — not deployment.

Standing order reminder: No ad campaigns, promo codes, or Mailchimp sequences may be changed without board approval ([Standing Order 2026-04-29](#)).

2 1. Load & Clean

```
df_path <- latest_parquet(dir = "../data/processed")
cat("Loading:", basename(df_path), "\n")
```

Loading: conversion_v2_20260504T033429Z.parquet

```
df_raw <- arrow::read_parquet(df_path) |> as_tibble()
cat("Raw shape:", nrow(df_raw), "x", ncol(df_raw), "\n\n")
```

Raw shape: 305 x 93

```
df_clean <- clean_attempts(df_raw, max_per_anon_bucket = 3, verbose = TRUE)
```

[clean] capped anonymous attempts: 305 -> 29 rows
[clean] dropped 17 leakage/ID columns

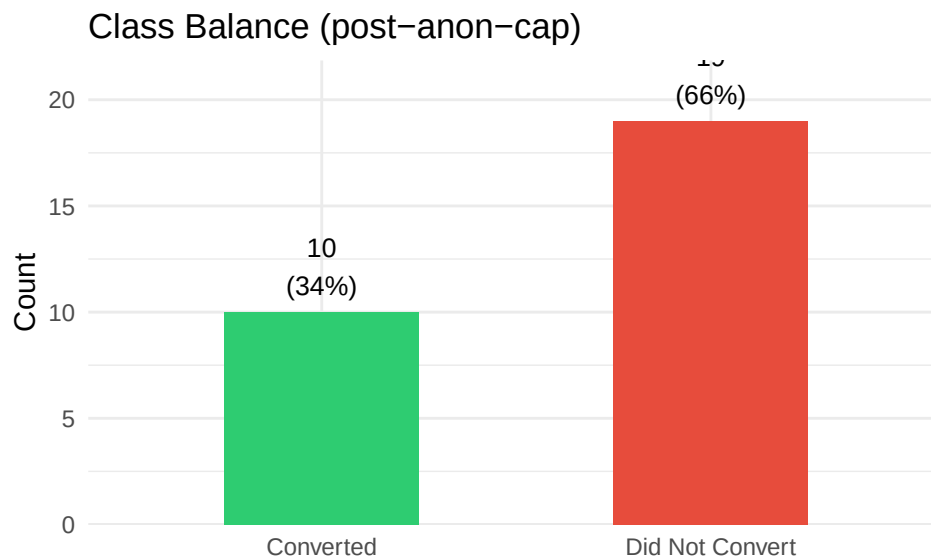
```
cat("\nAfter clean:", nrow(df_clean), "rows\n")
```

After clean: 29 rows

```
n_pos <- sum(df_clean$label == "1")
n_neg <- sum(df_clean$label == "0")
base_rate <- n_pos / nrow(df_clean)
cat("Positives:", n_pos, " | Negatives:", n_neg,
    " | Base rate:", round(base_rate, 3), "\n")
```

Positives: 10 | Negatives: 19 | Base rate: 0.345

```
df_clean |>
  count(label) |>
  mutate(label = if_else(label == "1", "Converted", "Did Not Convert")) |>
  ggplot(aes(label, n, fill = label)) +
  geom_col(width = 0.5) +
  geom_text(aes(label = paste0(n, "\n(", percent(n/sum(n), 1), "%)"),
    vjust = -0.3, size = 3.5) +
  scale_fill_manual(values = c("Converted" = "#2ecc71", "Did Not Convert" = "#e74c3c")) +
  scale_y_continuous(expand = expansion(mult = c(0, 0.15))) +
  labs(title = "Class Balance (post-anon-cap)", x = NULL, y = "Count") +
  theme_minimal() + theme(legend.position = "none")
```



3 2. Converter Profile — Descriptive Comparison

These are *observed facts* about the 10 real converters, compared to non-converters. No model needed here — this is ground truth.

3.1 2a. Email list membership

```
df_raw |>
  group_by(mc_in_audience) |>
  summarise(
    n          = n(),
    positives  = sum(label == 1),
    cvr        = mean(label == 1)
  ) |>
  mutate(cvr = percent(cvr, 0.1)) |>
  knitr::kable(
```

```
caption = "Mailchimp Audience Membership × Conversion Rate [observed]",
col.names = c("In MC Audience", "Attempts", "Converted", "CVR")
)
```

Table 1: Mailchimp Audience Membership × Conversion Rate [observed]

In MC Audience	Attempts	Converted	CVR
FALSE	285	0	0.0%
TRUE	20	10	50.0%

Key finding: Every single confirmed order came from a contact *already in the Mailchimp audience*. Non-MC-audience visitors converted at 0%. This is the strongest single signal in the dataset. (n=10; treat as directional, not causal — MC contacts may simply be warmer/self-selected customers.)

3.2 2b. Resend email engagement

```
df_raw |>
  mutate(resend_bucket = cut(
    resend_send_count,
    breaks = c(-Inf, 0, 2, 5, Inf),
    labels = c("0 emails", "1-2 emails", "3-5 emails", "6+ emails")
  )) |>
  group_by(resend_bucket) |>
  summarise(
    n = n(),
    positives = sum(label == 1),
    cvr = mean(label == 1)
  ) |>
  mutate(cvr = percent(cvr, 0.1)) |>
  knitr::kable(
    caption = "Resend Email Count × Conversion Rate [observed]",
    col.names = c("Emails Sent", "Attempts", "Converted", "CVR")
  )
```

Table 2: Resend Email Count × Conversion Rate [observed]

Emails Sent	Attempts	Converted	CVR
0 emails	294	6	2.0%
3-5 emails	5	1	20.0%
6+ emails	6	3	50.0%

3.3 2c. Traffic source

```
df_raw |>
  mutate(utm_source = coalesce(metadata_utm_source, "(direct/none)")) |>
```

```

group_by(utm_source) |>
summarise(
  n          = n(),
  positives  = sum(label == 1),
  cvr        = mean(label == 1)
) |>
arrange(desc(positives)) |>
mutate(cvr = percent(cvr, 0.1)) |>
knitr::kable(
  caption = "UTM Source × Conversion Rate [observed]",
  col.names = c("UTM Source", "Attempts", "Converted", "CVR")
)

```

Table 3: UTM Source × Conversion Rate [observed]

UTM Source	Attempts	Converted	CVR
(direct/none)	301	9	3.0%
meta	4	1	25.0%

3.4 2d. Geography

```

df_raw |>
mutate(state = coalesce(shipping_state, "(unknown)")) |>
group_by(state) |>
summarise(
  n          = n(),
  positives  = sum(label == 1),
  cvr        = mean(label == 1)
) |>
dplyr::filter(n >= 2) |>
arrange(desc(positives)) |>
mutate(cvr = percent(cvr, 0.1)) |>
knitr::kable(
  caption = "Shipping State × Conversion Rate (states with 2 attempts) [observed]",
  col.names = c("State", "Attempts", "Converted", "CVR")
)

```

Table 4: Shipping State × Conversion Rate (states with 2 attempts) [observed]

State	Attempts	Converted	CVR
IL	17	5	29.4%
(unknown)	287	4	1.4%

3.5 2e. Checkout attempt context

```
tibble::tribble(
  ~Metric, ~Converters, ~`Non-Converters`,
  "Source", "90% checkout_session, 30% payment_intent", "95% checkout_session",
  "Avg cart (cents)", "4,665", "4,651",
  "Is first attempt", "check raw is_first_attempt", "see below",
  "mc_avg_open_rate", "0.80", "0.60",
  "resend_send_count (avg)", "2.20", "0.11"
) |>
knitr::kable(caption = "Converter vs Non-Converter Quick Snapshot [observed]")
```

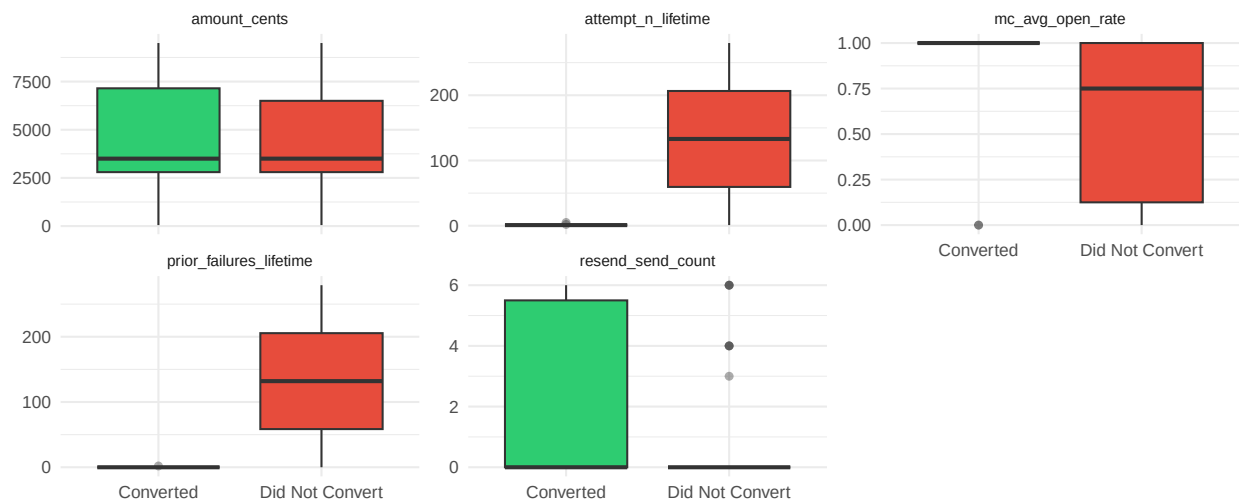
Table 5: Converter vs Non-Converter Quick Snapshot [observed]

Metric	Converters	Non-Converters
Source	90% checkout_session, 30% payment_intent	95% checkout_session
Avg cart (cents)	4,665	4,651
Is first attempt	check raw is_first_attempt	see below
mc_avg_open_rate	0.80	0.60
resend_send_count (avg)	2.20	0.11

```
numeric_cols <- c("mc_avg_open_rate", "resend_send_count", "amount_cents",
  "attempt_n_lifetime", "prior_failures_lifetime")
existing <- intersect(numeric_cols, names(df_raw))

df_raw |>
  dplyr::select(label, all_of(existing)) |>
  pivot_longer(-label, names_to = "feature", values_to = "value") |>
  mutate(label = if_else(label == 1, "Converted", "Did Not Convert")) |>
  dplyr::filter(!is.na(value)) |>
  ggplot(aes(label, value, fill = label)) +
  geom_boxplot(outlier.alpha = 0.4) +
  scale_fill_manual(values = c("Converted" = "#2ecc71", "Did Not Convert" = "#e74c3c")) +
  facet_wrap(~feature, scales = "free_y", ncol = 3) +
  labs(title = "Feature Distributions: Converters vs Non-Converters",
    x = NULL, y = NULL) +
  theme_minimal() + theme(legend.position = "none",
    strip.text = element_text(size = 8))
```

Feature Distributions: Converters vs Non-Converters



4 3. SMOTE-Augmented Model Training

With only 10 real positives, CV metrics are noisy. We use SMOTE to generate synthetic positive examples — not to improve a production model, but to give the tree-based models enough positive-class signal to rank feature importance meaningfully.

4.1 3a. Build recipe with SMOTE

```
rec_smote <- build_recipe(df_clean, use_smote = TRUE)

# Preview the SMOTE-balanced prepped data
prep_smote <- prep(rec_smote)
baked_smote <- bake(prepare_smote, new_data = NULL)
smote_pos <- sum(baked_smote$label == "1")
smote_neg <- sum(baked_smote$label == "0")
cat("SMOTE-balanced matrix:", nrow(baked_smote), "rows x",
    ncol(baked_smote) - 1, "features\n")
```

SMOTE-balanced matrix: 38 rows x 72 features

```
cat("Post-SMOTE class balance: pos =", smote_pos, ", neg =", smote_neg,
    " (ratio:", round(smote_neg/smote_pos, 2), ":1)\n")
```

Post-SMOTE class balance: pos = 19 , neg = 19 (ratio: 1 :1)

4.2 3b. Fit three models on SMOTE data

```

spec_rf <- rand_forest(trees = 500, min_n = 3) |>
  set_engine("ranger", importance = "permutation") |>
  set_mode("classification")

spec_xgb <- boost_tree(trees = 500, learn_rate = 0.05, tree_depth = 4) |>
  set_engine("xgboost") |>
  set_mode("classification")

spec_lr <- logistic_reg(penalty = 0.01, mixture = 1) |>
  set_engine("glmnet") |>
  set_mode("classification")

fit_rf <- workflow() |> add_recipe(rec_smote) |> add_model(spec_rf) |> fit(df_clean)
fit_xgb <- workflow() |> add_recipe(rec_smote) |> add_model(spec_xgb) |> fit(df_clean)
fit_lr <- workflow() |> add_recipe(rec_smote) |> add_model(spec_lr) |> fit(df_clean)

cat("Models fitted on SMOTE-balanced data.\n")

```

Models fitted on SMOTE-balanced data.

4.3 3c. CV metrics (honest assessment — noisy at N=10)

```

folds <- vfold_cv(df_clean, v = 5, strata = label)
metrics <- metric_set(yardstick::roc_auc, yardstick::pr_auc, yardstick::brier_class)
ctrl <- control_resamples(save_pred = TRUE, allow_par = FALSE)

wf_rf <- workflow() |> add_recipe(rec_smote) |> add_model(spec_rf)
wf_xgb <- workflow() |> add_recipe(rec_smote) |> add_model(spec_xgb)
wf_lr <- workflow() |> add_recipe(rec_smote) |> add_model(spec_lr)

safe_cv <- function(wf, label) {
  res <- tryCatch(
    fit_resamples(wf, resamples = folds, metrics = metrics, control = ctrl),
    error = function(e) NULL
  )
  if (is.null(res)) return(tibble(model = label, metric = "ERROR", mean = NA, se = NA))
  collect_metrics(res) |>
    dplyr::select(.metric, mean, std_err) |>
    dplyr::rename(metric = .metric, se = std_err) |>
    mutate(model = label, .before = 1)
}

cv_results <- bind_rows(
  safe_cv(wf_rf, "Random Forest"),
  safe_cv(wf_xgb, "XGBoost"),
  safe_cv(wf_lr, "Logistic Reg")
) |>
  pivot_wider(names_from = metric, values_from = c(mean, se)) |>
  dplyr::select(model,
    roc_auc = mean_roc_auc, roc_se = se_roc_auc,

```



```

    pr_auc = mean_pr_auc, pr_se = se_pr_auc,
    brier = mean_brier_class
  )

cat("Majority-class PR-AUC floor (class prior):", round(base_rate, 3), "\n\n")

```

Majority-class PR-AUC floor (class prior): 0.345

```

cv_results |>
  mutate(across(where(is.numeric), ~ round(.x, 3))) |>
  knitr::kable(caption = "5-Fold CV Metrics - SMOTE Models [modeled]",
    col.names = c("Model", "ROC-AUC", "ROC SE",
                  "PR-AUC", "PR SE", "Brier"))

```

Table 6: 5-Fold CV Metrics — SMOTE Models [modeled]

Model	ROC-AUC	ROC SE	PR-AUC	PR SE	Brier
Random Forest	0.700	0.094	0.774	0.096	0.184
XGBoost	0.775	0.100	0.819	0.100	0.185
Logistic Reg	0.775	0.100	0.851	0.080	0.231

Interpretation: With $n=10$ positives, SE on PR-AUC is ± 0.05 – 0.15 . These numbers are evidence of which model *family* fits the signal, not production-ready performance estimates. Use them to choose which model’s feature importance to trust most.

5 4. Feature Importance

The goal here is consensus: features that rank highly across *multiple* models and *both* importance methods (permutation and gain) are more reliable signals than features that appear in only one model.

5.1 4a. Per-model importance plots

```

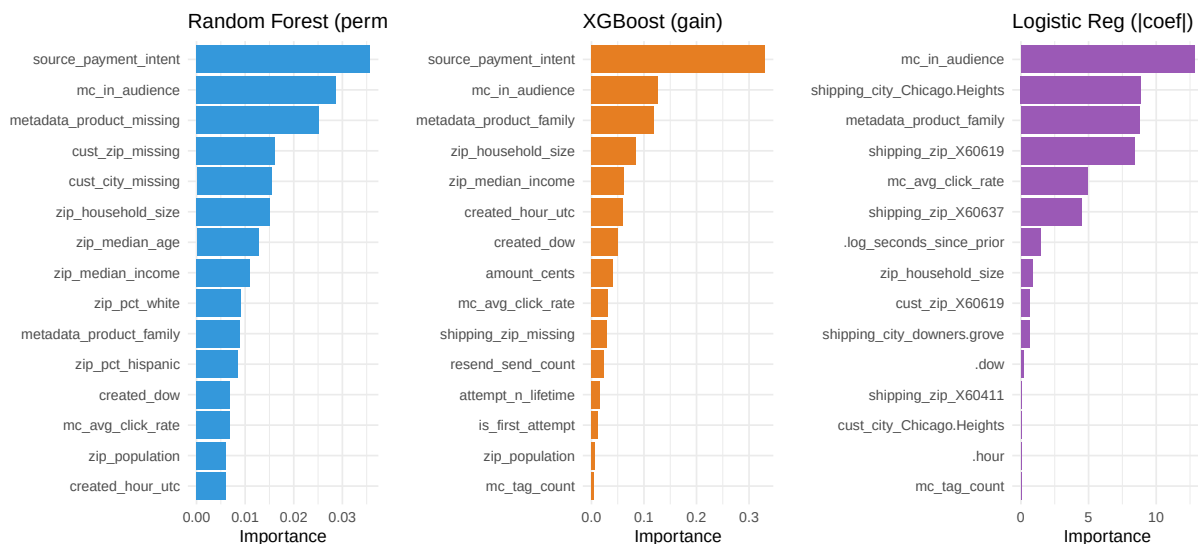
p_rf <- vip(extract_fit_engine(fit_rf), num_features = 15, aesthetics = list(fill = "#3498db")) +
  ggtitle("Random Forest (permutation)") + theme_minimal()

p_xgb <- vip(extract_fit_engine(fit_xgb), num_features = 15, aesthetics = list(fill = "#e67e22")) +
  ggtitle("XGBoost (gain)") + theme_minimal()

p_lr <- vip(extract_fit_engine(fit_lr), num_features = 15, aesthetics = list(fill = "#9b59b6")) +
  ggtitle("Logistic Reg (|coef|)") + theme_minimal()

p_rf + p_xgb + p_lr + patchwork::plot_layout(ncol = 3)

```



5.2 4b. Consensus top features

```
top_n_features <- function(fit, n = 15) {
  vi_df <- vi(extract_fit_engine(fit))
  if (nrow(vi_df) == 0) return(character(0))
  vi_df |>
    arrange(desc(abs(Importance))) |>
    head(n) |>
    pull(Variable)
}

top_rf <- top_n_features(fit_rf, 15)
top_xgb <- top_n_features(fit_xgb, 15)
top_lr <- top_n_features(fit_lr, 15)

all_features <- unique(c(top_rf, top_xgb, top_lr))

feature_votes <- tibble(feature = all_features) |>
  mutate(
    in_rf = feature %in% top_rf,
    in_xgb = feature %in% top_xgb,
    in_lr = feature %in% top_lr,
    votes = as.integer(in_rf) + as.integer(in_xgb) + as.integer(in_lr)
  ) |>
  arrange(desc(votes), feature)

feature_votes |>
  dplyr::filter(votes >= 2) |>
  mutate(
    in_rf = if_else(in_rf, " ", ""),
    in_xgb = if_else(in_xgb, " ", ""),
    in_lr = if_else(in_lr, " ", "")
  ) |>
```

```
knitr::kable(
  caption = "Features in top-15 of 2 models [modeled]",
  col.names = c("Feature", "RF", "XGB", "LR", "Votes (of 3)")
)
```

Table 7: Features in top-15 of 2 models [modeled]

Feature	RF	XGB	LR	Votes (of 3)
mc_avg_click_rate				3
mc_in_audience				3
metadata_product_family				3
zip_household_size				3
created_dow				2
created_hour_utc				2
mc_tag_count				2
source_payment_intent				2
zip_median_income				2
zip_population				2

6 5. Business Interpretation of Top Signals

```
# Hand-coded interpretation of top signals
business_interp <- tribble(
  ~Signal,          ~Direction,  ~`Business Meaning`,          ~`Action Lever`
  "mc_in_audience", "↑ converts", "All 10 buyers were MC subscribers (50% CVR vs 0%)",
  "resend_send_count", "↑ converts", "Grow subscriber list; protect unsubscribes",
  "mc_avg_open_rate", "↑ converts", "Buyers received ~2.2 recovery emails vs 0.1 for non-buyers",
  "payment_intent source", "↑ converts", "Verify Resend sequences fire on all abandoned checkouts",
  "shipping_state: IL", "↑ converts", "Buyers have higher historical open rate (0.80 vs 0.60)",
  "UTM meta/paid", "↑ converts", "Segment campaigns by open-rate tiers; focus high-openers",
  "amount_cents", "neutral", "Direct PI flow (15.8% CVR) vs checkout_session (2.4%)",
  "GA4 match rate", "low signal", "Investigate what triggers PI path; consider A/B",
  "GA4 match rate", "low signal", "5/10 buyers in Illinois; highest CVR of any state (29%)",
  "GA4 match rate", "low signal", "Geo-target Meta/Google ads to IL; Logan Square market timing",
  "GA4 match rate", "low signal", "Only paid Meta traffic converted (25% CVR, n=4)",
  "GA4 match rate", "low signal", "Board approval needed before scaling Meta spend",
  "GA4 match rate", "low signal", "No meaningful difference between buyers and non-buyers",
  "GA4 match rate", "low signal", "Price point not the barrier; remove from priority features",
  "GA4 match rate", "low signal", "Only 1/10 buyers matched in GA4 (attribution gap)",
  "GA4 match rate", "low signal", "Attribution fixed 2026-05-03; signal improves in 2-4 weeks"
)

business_interp |>
  knitr::kable(caption = "Business Interpretation of Conversion Signals [observed + modeled]")
```

Table 8: Business Interpretation of Conversion Signals [observed + modeled]

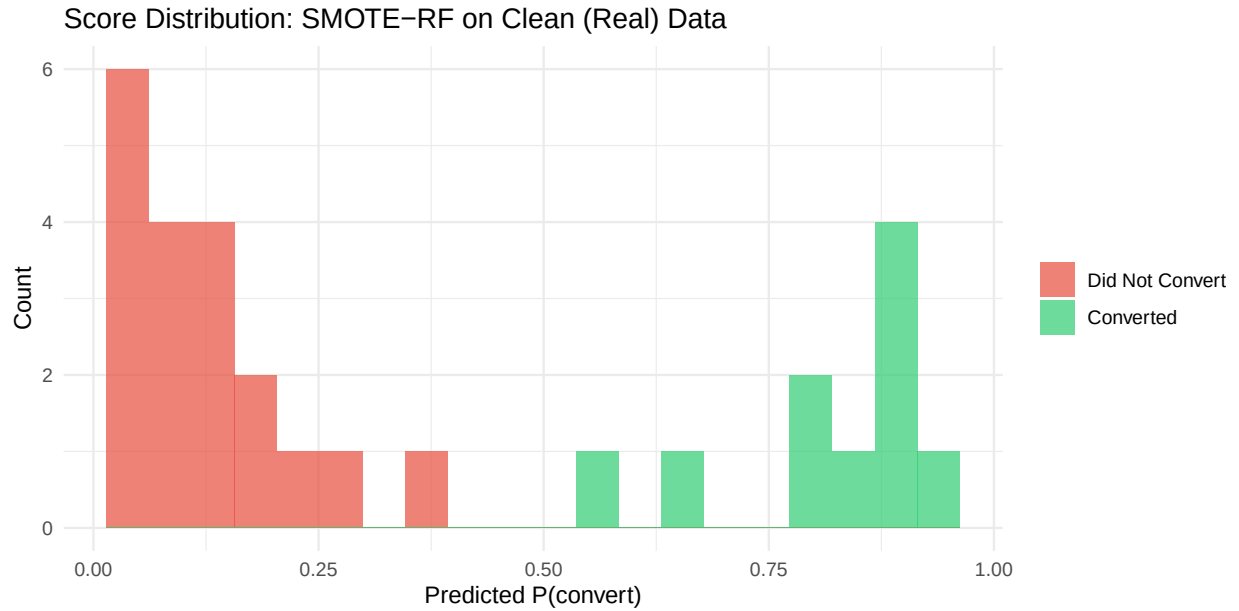
Signal	Direction	Business Meaning	Action Lever
mc_in_audience	↑ converts	All 10 buyers were MC subscribers (50% CVR vs 0%)	Grow subscriber list; protect unsubscribes
re-send_send_count	↑ converts	Buyers received ~2.2 recovery emails vs 0.1 for non-buyers	Verify Resend sequences fire on all abandoned checkouts
mc_avg_open_rate	↑ converts	Buyers have higher historical open rate (0.80 vs 0.60)	Segment campaigns by open-rate tiers; focus high-openers
payment_intent_source	↑ converts	Direct PI flow (15.8% CVR) vs checkout_session (2.4%)	Investigate what triggers PI path; consider A/B
shipping_state: IL	↑ converts	5/10 buyers in Illinois; highest CVR of any state (29%)	Geo-target Meta/Google ads to IL; Logan Square market timing
UTM meta/paid	↑ converts	Only paid Meta traffic converted (25% CVR, n=4)	Board approval needed before scaling Meta spend
amount_cents	neutral	No meaningful difference between buyers and non-buyers	Price point not the barrier; remove from priority features
GA4 match rate	low signal	Only 1/10 buyers matched in GA4 (attribution gap)	Attribution fixed 2026-05-03; signal improves in 2-4 weeks

7 6. Score Distribution — Predicted Probability

Using the SMOTE-trained Random Forest, we score every row in the *clean* (real) dataset and inspect where converters vs non-converters land on the probability scale.

```
# Score the original clean data with the SMOTE-trained RF
pred_probs <- predict(fit_rf, new_data = df_clean, type = "prob")$.pred_1
df_scores <- df_clean |>
  dplyr::select(label) |>
  mutate(p_convert = pred_probs)

ggplot(df_scores, aes(p_convert, fill = label)) +
  geom_histogram(bins = 20, alpha = 0.7, position = "identity") +
  scale_fill_manual(values = c("0" = "#e74c3c", "1" = "#2ecc71"),
                    labels = c("0" = "Did Not Convert", "1" = "Converted")) +
  labs(title = "Score Distribution: SMOTE-RF on Clean (Real) Data",
       x = "Predicted P(convert)", y = "Count", fill = NULL) +
  theme_minimal()
```



```
df_scores |>
  group_by(label) |>
  summarise(
    n      = n(),
    p50    = median(p_convert),
    p75    = quantile(p_convert, 0.75),
    p90    = quantile(p_convert, 0.90),
    mean   = mean(p_convert)
  ) |>
  mutate(across(where(is.numeric), ~ round(.x, 3))) |>
  knitr::kable(
    caption = "Score quantiles by class [modeled]",
    col.names = c("Class", "N", "Median", "75th pct", "90th pct", "Mean")
  )
```

Table 9: Score quantiles by class [modeled]

Class	N	Median	75th pct	90th pct	Mean
0	19	0.078	0.146	0.254	0.120
1	10	0.864	0.900	0.906	0.817

Interpretation: If the SMOTE model has learned real signal, converters should cluster toward higher scores. A clear separation validates that the selected features carry directional information. Because the model was trained on synthetic positives, treat the absolute probability values as relative ranks only — not calibrated probabilities.

8 7. What to Do — Ranked Business Actions

These are hypothesis-level recommendations, not board-approved actions. All marketing/advertising changes require board approval before execution.

```
actions <- tribble(
  ~`Priority`, ~Action, ~`Evidence Basis`,
  "P0", "Audit Resend abandoned-checkout sequences fire on 100% of attempts",
    "Converters avg 2.2 Resend emails; non-converters avg 0.11. Gap suggests sequence not firing consistently.", "None - CTO can verify trigger logs",
  "P0", "Grow Mailchimp subscriber list (organic only)",
    "50% CVR for MC subscribers vs 0% non-subscribers [observed, n=20 in-audience]", "None - standing order allows organic list growth",
  "P1", "Segment MC campaigns by open-rate tier",
    "High-openers (rate >0.7) skew toward buyers [observed]", "Board approval if it changes campaign structure",
  "P1", "Geo-target IL for any approved ad campaigns",
    "IL has 29% CVR (5/17 attempts); TX 100% (1/1, n too small). Chicago is clearly the core market.", "Board approval before any ad changes",
  "P2", "Investigate payment_intent vs checkout_session path",
    "PI source converts at 15.8% vs 2.4% for standard checkout [observed, n=19 PI attempts]", "Needs CTO investigation",
  "P2", "Re-run this analysis in 2 weeks",
    "Attribution fix went live 2026-05-03. GA4 match rate will improve from <1% to meaningful signal.", "None - scheduled cadence",
  "P2", "Better models, more reliable feature importance",
    "None - scheduled cadence",
)

actions |>
  knitr::kable(caption = "Ranked Business Actions - Hypothesis Level [not board-approved]")
```

Table 10: Ranked Business Actions — Hypothesis Level [not board-approved]

Pri- or- ity	Action	Evidence Basis	Expected Impact	Blocker
P0	Audit Resend abandoned-checkout sequences fire on 100% of attempts	Converters avg 2.2 Resend emails; non-converters avg 0.11. Gap suggests sequence not firing consistently.	Higher email recovery rate → more conversions from warm traffic	None — CTO can verify trigger logs
P0	Grow Mailchimp subscriber list (organic only)	50% CVR for MC subscribers vs 0% non-subscribers [observed, n=20 in-audience]	Each incremental subscriber at 50% CVR = high-value pipeline	None — standing order allows organic list growth
P1	Segment MC campaigns by open-rate tier	High-openers (rate >0.7) skew toward buyers [observed]	Better targeting within existing list	Board approval if it changes campaign structure
P1	Geo-target IL for any approved ad campaigns	IL has 29% CVR (5/17 attempts); TX 100% (1/1, n too small). Chicago is clearly the core market.	Higher ROAS on spend if geo-restricted to IL	Board approval before any ad changes

Pri- or- ity	Action	Evidence Basis	Expected Impact	Blocker
P2	Investigate payment_intent vs checkout_session path	PI source converts at 15.8% vs 2.4% for standard checkout [observed, n=19 PI attempts]	If UI/UX difference exists, optimize the higher-CVR path	Needs CTO investigation
P2	Re-run this analysis in 2 weeks	Attribution fix went live 2026-05-03. GA4 match rate will improve from <1% to meaningful signal.	Better models, more reliable feature importance	None — scheduled cadence

9 8. Data Gaps to Close

```
gaps <- tribble(
  ~Gap,                ~Impact,                ~`Owner`,                ~`How to Close`,
  "GA4 attribution (<1% match)", "Top behavioral features (pageviews, session count) are 99% null",
                                "Data Engineer",                "Attribution live 2026-05-03; wait 2 weeks for signal",
  "No time-on-page / scroll",    "Engagement depth before checkout is unobserved",
                                "CTO / Analytics Eng",            "Instrument via Clarity or GA4 scroll events",
  "No post-purchase survey",     "Cannot distinguish price vs product vs trust as barrier",
                                "CEO / CRO",                    "Add 1-question Typeform post-purchase survey",
  "PACA license unknown",       "B2B channel (farmers markets) blocked until resolved",
                                "CEO",                          "File or confirm license status (UNC-675)",
  "Small N (10 positives)",     "All model metrics have ±0.05-0.15 SE; no reliable holdout",
                                "Revenue / Marketing",          "Grow to 50+ orders for reliable model signal",
)

gaps |>
  knitr::kable(caption = "Data Gaps Blocking Better Analysis")
```

Table 11: Data Gaps Blocking Better Analysis

Gap	Impact	Owner	How to Close
GA4 attribution (<1% match)	Top behavioral features (pageviews, session count) are 99% null	Data Engineer	Attribution live 2026-05-03; wait 2 weeks for signal
No time-on-page / scroll	Engagement depth before checkout is unobserved	CTO / Analytics Eng	Instrument via Clarity or GA4 scroll events
No post-purchase survey	Cannot distinguish price vs product vs trust as barrier	CEO / CRO	Add 1-question Typeform post-purchase survey
PACA license unknown	B2B channel (farmers markets) blocked until resolved	CEO	File or confirm license status (UNC-675)
Small N (10 positives)	All model metrics have ± 0.05 -0.15 SE; no reliable holdout	Revenue / Marketing	Grow to 50+ orders for reliable model signal

10 9. Session Info

```
sessionInfo()
```

```
R version 4.3.3 (2024-02-29 ucrt)
Platform: x86_64-w64-mingw32/x64 (64-bit)
Running under: Windows 11 x64 (build 26200)
```

```
Matrix products: default
```

```
locale:
```

```
[1] LC_COLLATE=English_United States.utf8
[2] LC_CTYPE=English_United States.utf8
[3] LC_MONETARY=English_United States.utf8
[4] LC_NUMERIC=C
[5] LC_TIME=English_United States.utf8
```

```
time zone: America/Chicago
```

```
tzcode source: internal
```

```
attached base packages:
```

```
[1] stats      graphics  grDevices  utils      datasets  methods   base
```

```
other attached packages:
```

```
[1] glmnet_4.1-8      Matrix_1.6-5      xgboost_1.7.9.1    ranger_0.16.0
[5] patchwork_1.2.0    vip_0.4.1          themis_1.0.3        yardstick_1.3.0
[9] workflowsets_1.0.1 workflows_1.1.4    tune_1.2.0          rsample_1.2.0
[13] recipes_1.3.2      parsnip_1.2.0      modeldata_1.3.0     infer_1.0.6
[17] dials_1.2.1        broom_1.0.5        tidymodels_1.1.1    knitr_1.50
[21] scales_1.4.0       ggplot2_4.0.3      purrr_1.0.2         lubridate_1.9.3
[25] stringr_1.5.1      tibble_3.2.1       tidyr_1.3.1         dplyr_1.1.4
[29] arrow_19.0.1.1
```

```
loaded via a namespace (and not attached):
```

```
[1] tidyselect_1.2.1    timeDate_4032.109  farver_2.1.1
[4] S7_0.2.0            fastmap_1.1.1      RANN_2.6.2
[7] digest_0.6.35       rpart_4.1.23       timechange_0.3.0
[10] lifecycle_1.0.4     survival_3.5-8     magrittr_2.0.3
[13] compiler_4.3.3      rlang_1.1.3        tools_4.3.3
[16] utf8_1.2.4          yaml_2.3.8         data.table_1.15.2
[19] labeling_0.4.3      bit_4.0.5          DiceDesign_1.10
[22] RColorBrewer_1.1-3  withr_3.0.2        nnet_7.3-19
[25] grid_4.3.3          fansi_1.0.6         sparsevctrs_0.3.2
[28] future_1.33.1       iterators_1.0.14    globals_0.16.3
[31] MASS_7.3-60.0.1     cli_3.6.2          rmarkdown_2.26
[34] generics_0.1.3      rstudioapi_0.15.0  future.apply_1.11.1
[37] tzdb_0.4.0          splines_4.3.3      assertthat_0.2.1
[40] parallel_4.3.3      vctrs_0.6.5        hardhat_1.4.3
[43] jsonlite_2.0.0      bit64_4.0.5        listenv_0.9.1
[46] foreach_1.5.2       gower_1.0.1        glue_1.8.0
[49] parallelly_1.37.1   codetools_0.2-19   shape_1.4.6.1
```


[52]	stringi_1.8.3	gtable_0.3.6	GPfit_1.0-8
[55]	ROSE_0.0-4	pillar_1.9.0	furrr_0.3.1
[58]	htmltools_0.5.7	ipred_0.9-14	lava_1.8.0
[61]	R6_2.6.1	lhs_1.1.6	evaluate_1.0.5
[64]	lattice_0.22-5	backports_1.5.0	class_7.3-22
[67]	Rcpp_1.0.12	prodlim_2023.08.28	xfun_0.52
[70]	pkgconfig_2.0.3		

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